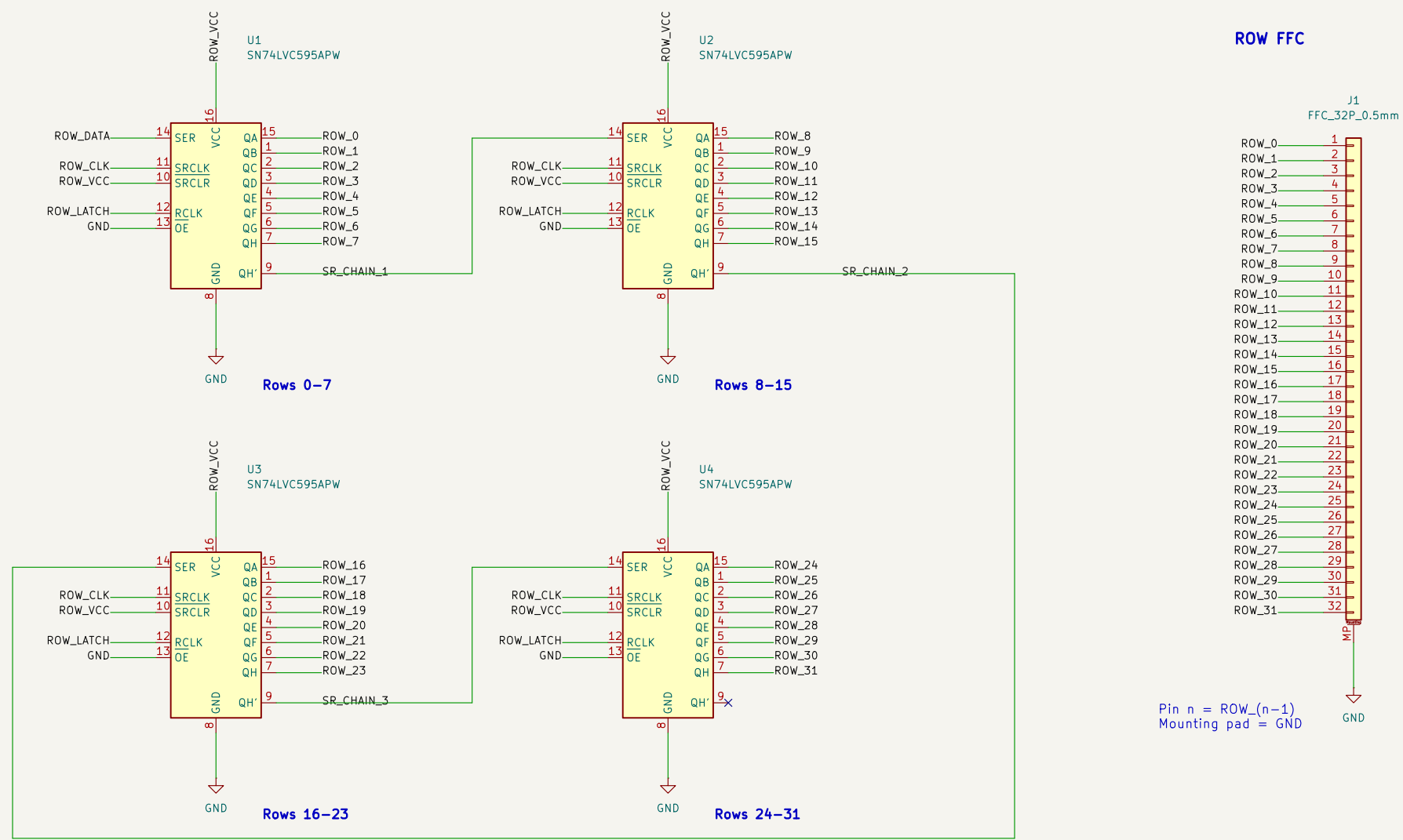
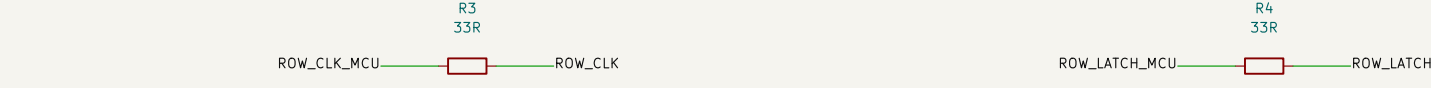


01 ROW EXCITATION

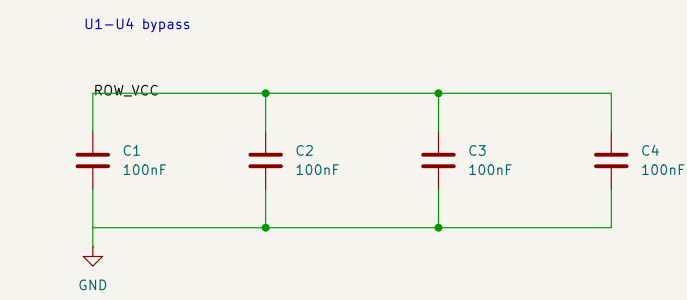
MCU serial stream > four latched drivers > 32-way row FFC



Serial order: U1 > U2 > U3 > U4
Outputs enabled; unselected rows held LOW.



Clock / latch damping

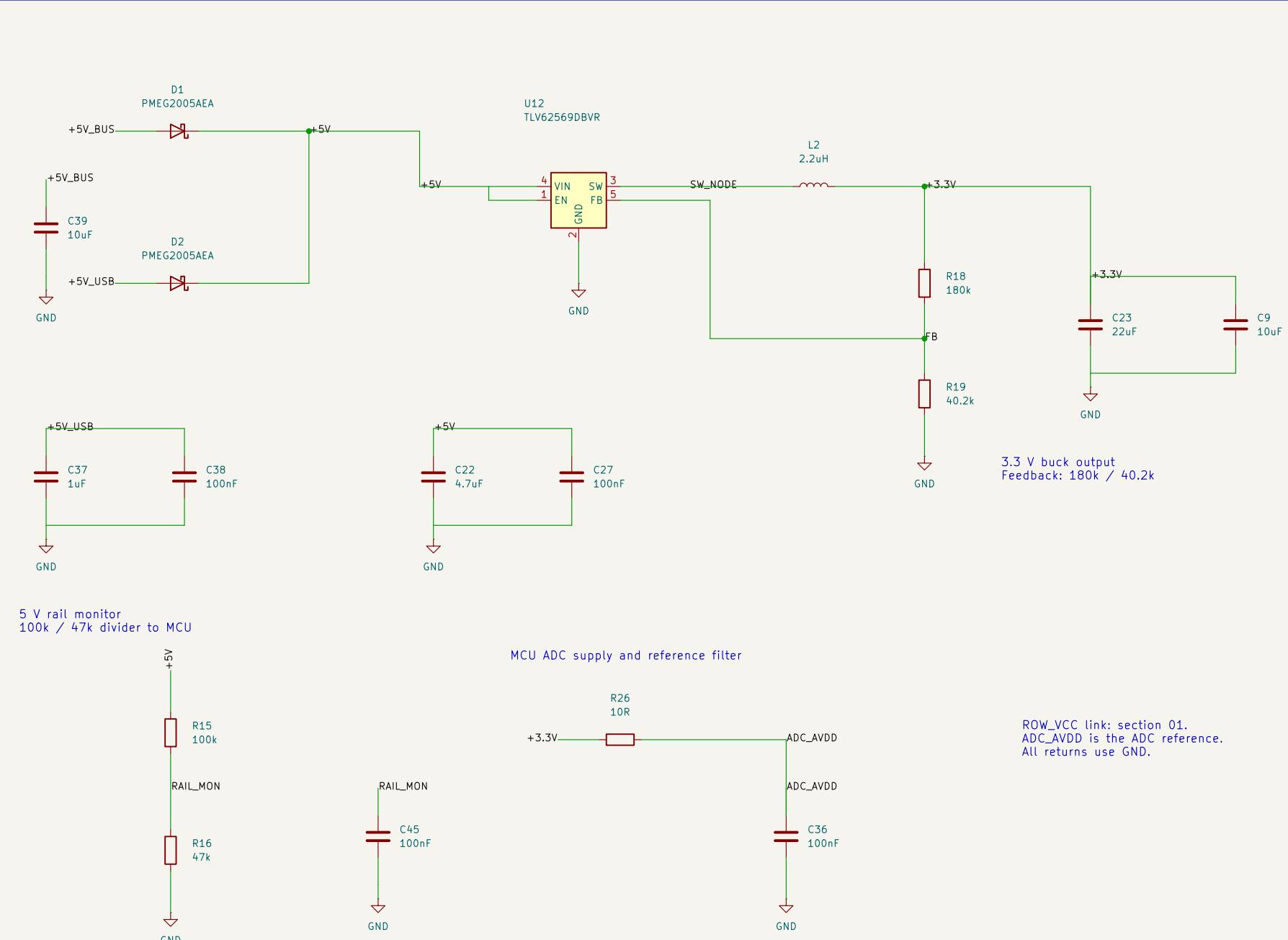


Excitation rail link



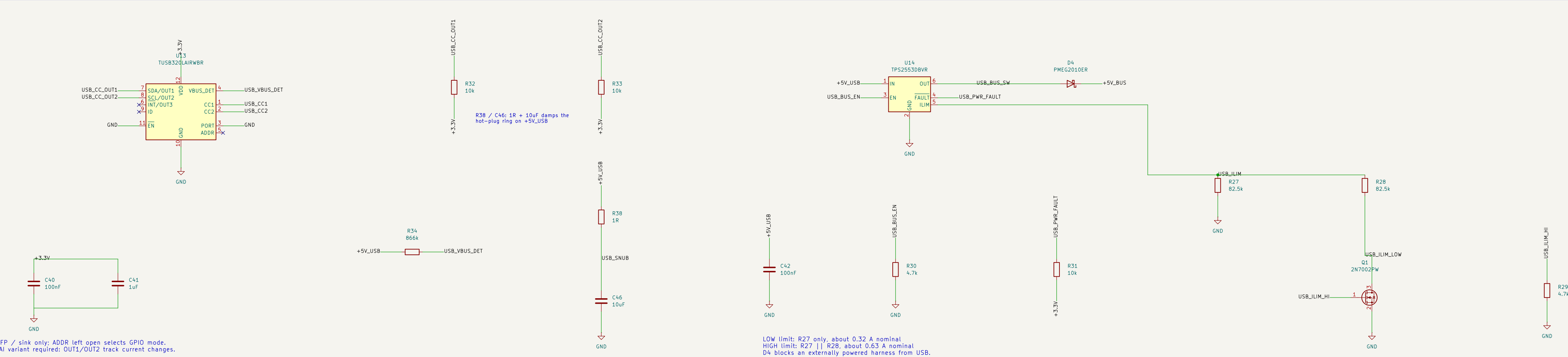
04 POWER + RAIL MONITOR

Harness or USB power > diode OR > local 3.3 V buck

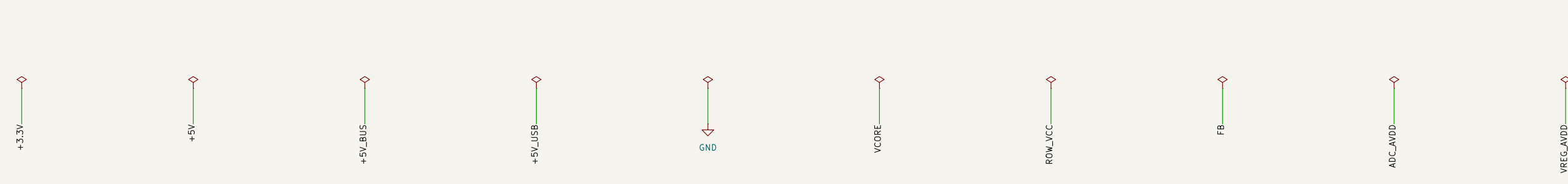


07 USB-C POWER TO THE HARNESS

Computer port current detection > firmware authorization > current limit > reverse barrier > +5V_BUS



POWER-NET DRIVE DECLARATIONS

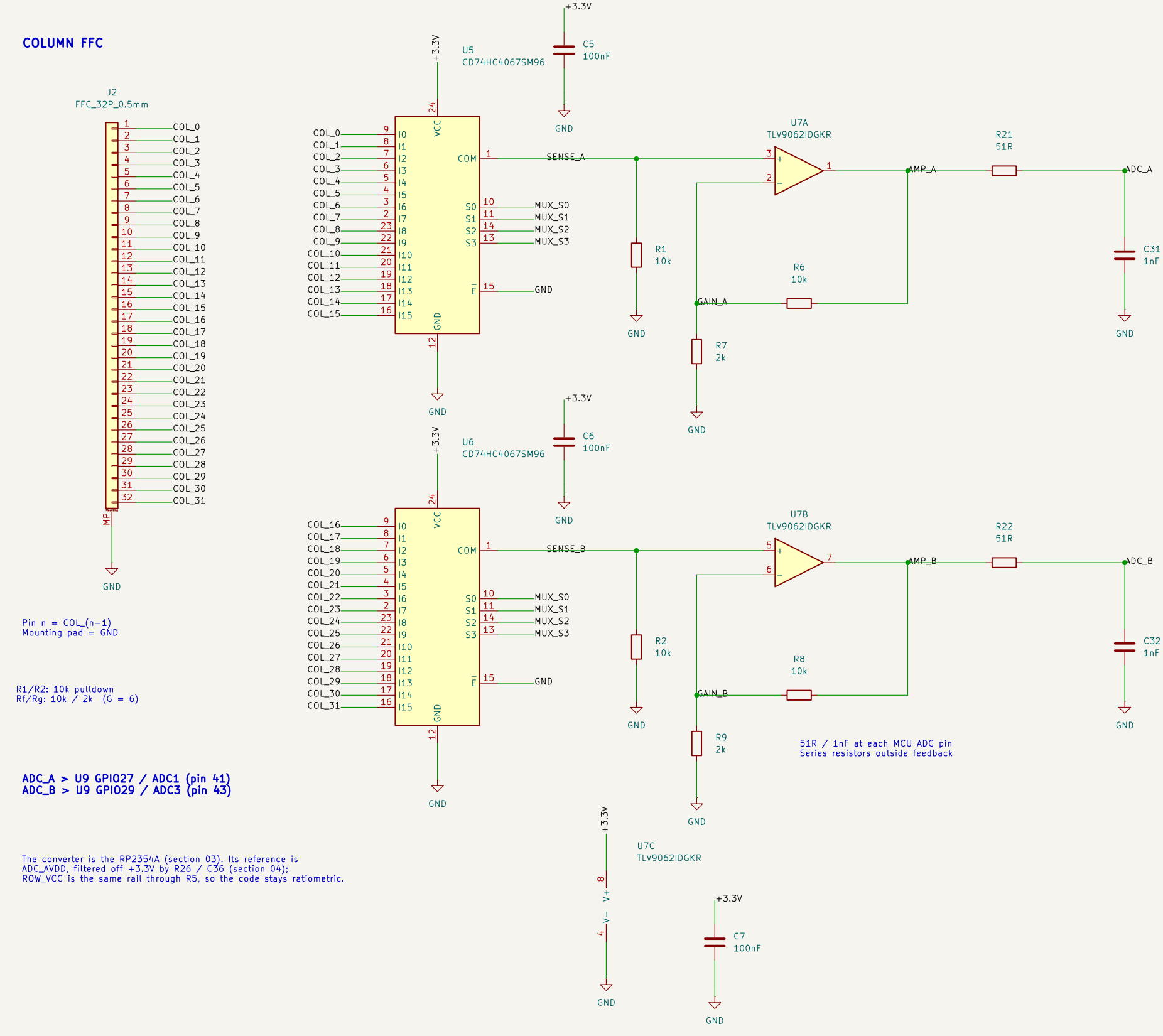


Sensor path: ROW_n > mat element > COL_n > SENSE_A/B > AMP_A/B > ADC_A/B

R1/R2 and gain-network values remain fit-on-test values from the existing design.

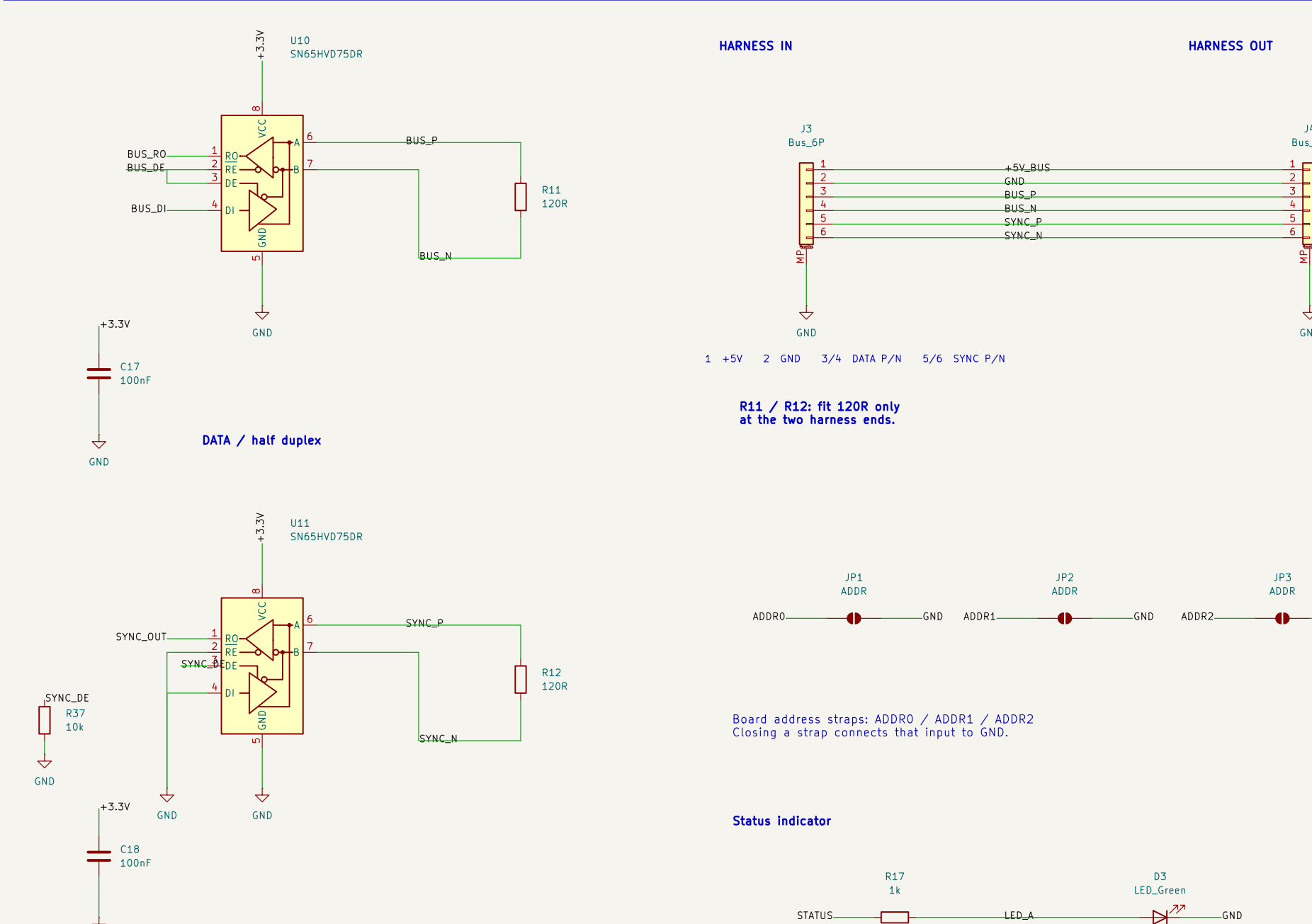
02 COLUMN SENSING

32 columns > two 16:1 muxes > gain of 6 > filtered inputs to the MCU ADC



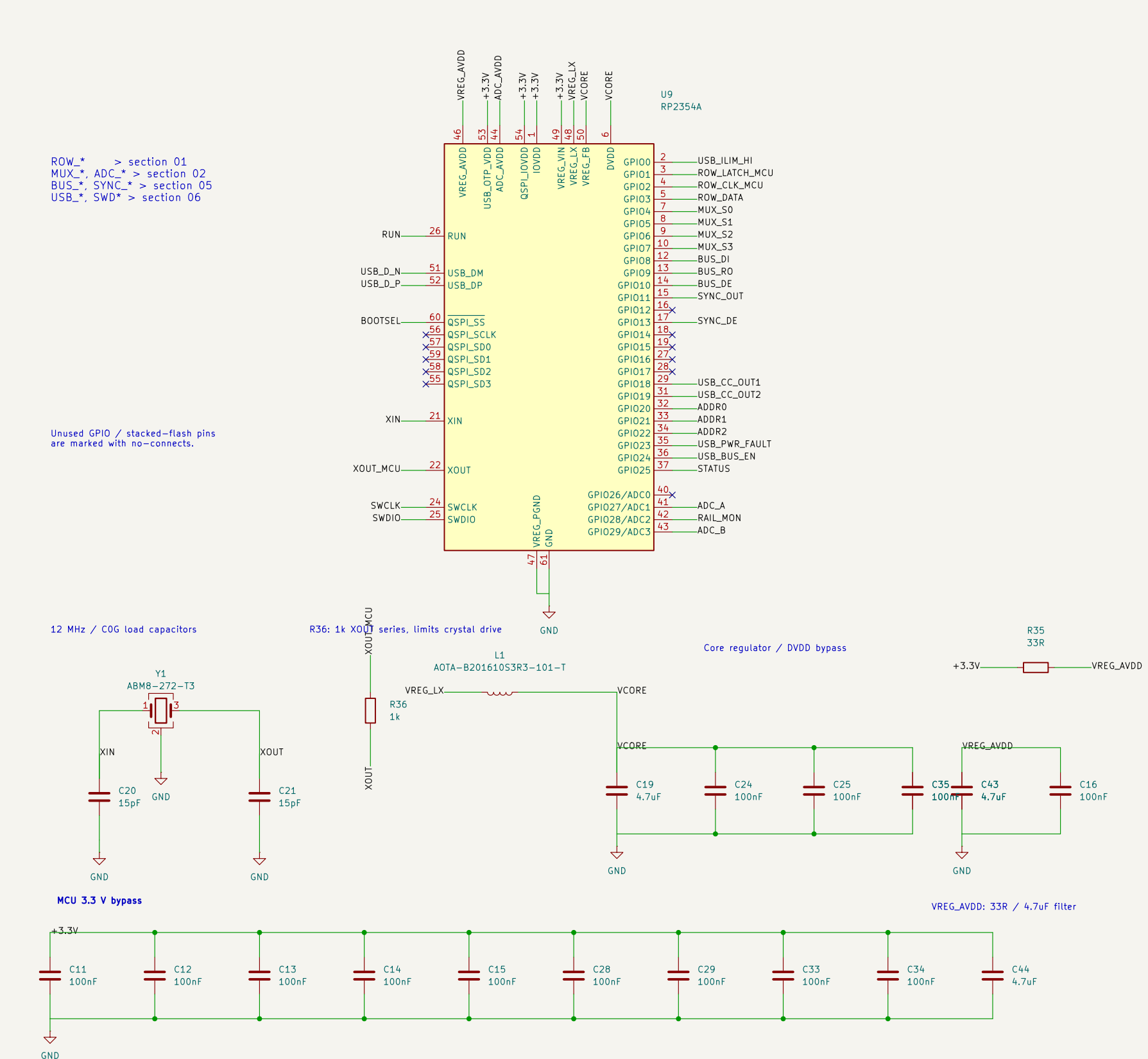
05 COMMUNICATIONS + BOARD ADDRESS

Half-duplex data pair and dedicated receive-only sync pair



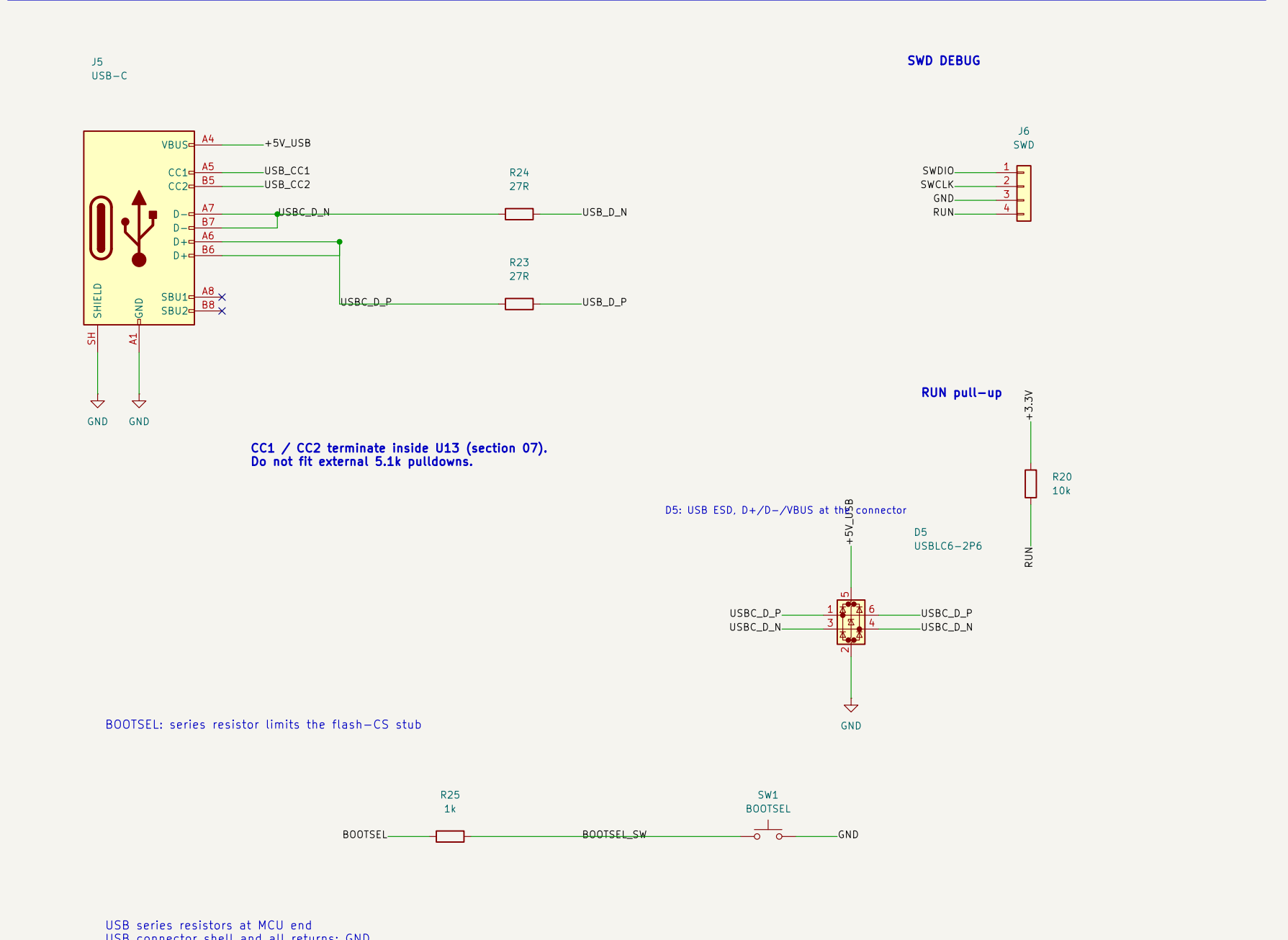
03 SCAN CONTROLLER

RP2354A / clock / core supply / local bypass



06 USB + PROGRAMMING

USB-C data / configuration straps / SWD / BOOTSEL



07 USB-C POWER TO THE HARNESS

Computer port current detection > firmware authorization > current limit > reverse barrier > +5V_BUS



POWER-NET DRIVE DECLARATIONS



Sensor path: ROW_n > mat element > COL_n > SENSE_A/B > AMP_A/B > ADC_A/B

R1/R2 and gain-network values remain fit-on-test values from the existing design.

Reading the sheet

Matching net labels connect between sections. A dot marks a branch; an X marks an unused pin.
Bypass capacitors are grouped by supply. USB attach current and the complete eight-board startup / operating budget require measurements; schematic checks cannot establish them.